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Ground shields positioned between columns of contacts

Abstract

An electrical connector which controls cross talk and signal radiation. The electrical connector includes columns of contacts having signal contacts and ground contacts. Ground shields are positioned between the columns of contacts. The ground shields have planar members with mating ends and mounting ends. The planar members have surfaces which are configured to be electrically coupled to ground members on a mating connector. Voids are provided in the planar members. Mounting members extend from the mounting ends of the planar members. The mounting members are configured to cooperate with the ground contacts provided in the columns of contacts.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to ground shields positioned between columns of contacts to reduce crosstalk for improved signal integrity. The ground connection with the ground shields can be made through direct physical contact, through capacitive coupling or through both.

BACKGROUND OF THE INVENTION

(2) Due to the increasing complexity of electronic components, it is desirable to fit more components in less space on a circuit board or other substrate. Consequently, the spacing between electrical terminals within connectors has been reduced, while the number of electrical terminals housed in the connectors has increased, thereby increasing the need in the electrical arts for electrical connectors that are capable of handling higher and higher speeds and to do so with greater and greater pin densities. It is desirable for such connectors to have not only reasonably constant impedance levels, but also acceptable levels of impedance and crosstalk, as well as other acceptable electrical and mechanical characteristics. Therefore, there remains a need to provide appropriate shielding to preserve signal integrity and to minimize crosstalk as speeds of signals increase and the footprint of the connector remains the same or decreases.

(3) It would, therefore, be beneficial to provide a connector ground shields positioned between columns of contacts, and in particular between signal pairs of contacts, to optimize shielding and reduce crosstalk for improved signal integrity. It would also be beneficial if the ground connection with the ground shields could be made through direct physical contact, capacitive coupling or both.

SUMMARY OF THE INVENTION

(4) An embodiment is directed to an electrical connector which controls cross talk and signal radiation. The electrical connector includes columns of contacts having signal contacts and ground contacts. Ground shields are positioned between the columns of contacts. The ground shields have planar members with mating ends and mounting ends. The planar members have surfaces which are configured to be electrically coupled to ground members on a mating connector. Voids are provided in the planar members. Mounting members extend from the mounting ends of the planar members. The mounting members are configured to cooperate with the ground contacts provided in the columns of contacts.

(5) An embodiment is directed to an electrical connector with spaced columns of contacts. Each respective row of the columns of contacts have signal contacts and ground contacts. Ground shields are positioned between the columns of contacts. The ground shields have planar members having mating ends and mounting ends. The planar members have surfaces which are configured to be electrically coupled to ground members on a mating connector. Voids are provided in the planar members. Mounting members extend from the mounting ends of the planar members. The mounting members have contact receiving openings through which the ground contacts extend. The mounting members are positioned in line with the voids of the planar members.

(6) Other features and advantages of the present invention will be apparent from the following more detailed description of the illustrative embodiment, taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a bottom, front perspective view of an illustrative connector system shown in an assembled position, with a backplane connector mated to a mating daughtercard connector.
- (2) FIG. 2 is a bottom, back perspective view of an the illustrative first backplane connector of FIG. 1.
- (3) FIG. 3 is an enlarged cross-sectional view of the daughtercard connector mated with the backplane connector, taken along line 3-3 of FIG. 1.
- (4) FIG. 4 is an enlarged cross-sectional view of the daughtercard connector mated with the backplane connector, taken along line 4-4 of FIG. 1.
- (5) FIG. 5 is a perspective view of two ground planes or shields of FIG. 1.
- (6) FIG. 6 is a perspective view of two ground planes or shields of an illustrative alternate second backplane connector.
- (7) FIG. 7 is an enlarged cross-sectional view of a daughtercard connector mated with the second backplane connector, with an orientation similar to that of FIG. 3.
- (8) FIG. 8 is an enlarged cross-sectional view of the daughtercard connector mated with the second backplane connector, with an orientation similar to that of FIG. 4.
- (9) FIG. 9 is a perspective view of two ground planes or shields of an illustrative alternate third backplane connector.
- (10) FIG. 10 is a bottom, back perspective view of an the illustrative third backplane connector of FIG. 9.

DETAILED DESCRIPTION OF THE INVENTION

- (11) The description of illustrative embodiments according to principles of the present invention is

intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top” and “bottom” as well as derivative thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise.

(12) Moreover, the features and benefits of the invention are illustrated by reference to the preferred embodiments. Accordingly, the invention expressly should not be limited to such embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features, the scope of the invention being defined by the claims appended hereto.

(13) FIG. 1 illustrates an illustrative electrical connector system **10** having a backplane connector **12** and a daughtercard connector **14** that are used to electrically connect a backplane circuit board (not shown) and a daughtercard circuit board (not shown). While the electrical connector system **10** is described herein with reference to backplane connectors **12** and daughtercard connectors **14**, it is realized that the subject matter herein may be utilized with different types of electrical connectors other than a backplane connector or a daughtercard connector. The backplane connector **12**, and the daughtercard connector **14** are merely illustrative of an electrical connector system **10**.

(14) In the illustrative embodiment shown, the daughtercard connector **14** constitutes a right angle connector wherein a mating interface **16** and mounting interface **18** of the daughtercard connector **14** are oriented perpendicular to one another. The daughtercard connector **14** is mounted to the daughtercard circuit board at the mounting interface **18**. Other orientations of the interfaces **16**, **18** are possible in alternative embodiments.

(15) The daughtercard connector **14** includes a housing **20**, made of one or more components, holding a plurality of circuit boards **22** therein. The circuit boards **22** have pairs of individual signal pathways or traces (not shown) that extend between the mating interface **16** and the mounting interface **18**. The signal traces have signal conductive pads **24** (FIG. 4) provided proximate the mating interface **16**. The circuit boards **22** have individual ground pathways or traces (not shown) that extend between the mating interface **16** and the mounting interface **18**. The ground traces have ground conductive pads **26** (FIG. 4) provided proximate the mating interface **16**. The circuit boards **22** also have ground pathways or traces **28** (FIG. 1) on opposite sides of the circuit boards **22** from the ground traces. The ground traces **28** extend between the mating interface **16** and the mounting interface **18**. In alternative embodiments, the circuit boards **22** may be contact modules, the signal traces may be mating signal contacts and the ground traces may be ground contacts.

(16) In the illustrated embodiment, the backplane connector **12** constitutes a header connector mounted to the backplane circuit board. However, other types of connectors may be used. When the backplane connector **12** is mated to the daughtercard connector **14**, the daughtercard circuit board is oriented generally perpendicular with respect to the backplane circuit board.

(17) As is shown in FIG. 2, the backplane connector **12** includes a mating end **30** and a mounting end **32** that are oriented generally parallel to one another. The backplane connector **12** is mounted to the backplane circuit board at the mounting end **32**. Other orientations of the mating end **30** and the mounting end **32** are possible in alternative embodiments.

(18) The backplane connector **12** includes a one piece housing **34**. Alternatively, the housing may

include a plurality of individual housings or modules. The housing **34** holds a at least one row of contacts **40**, which include signal contacts **42** and ground contacts **44**.

(19) The signal contacts **42** extend between the mating end **30** and the mounting end **32**. In the exemplary embodiment, the signal contacts **42** are arranged in pairs carrying differential signals. However, in other configurations, the signal contacts **42** may not be arranged in pairs for carrying differential signals.

(20) The ground contacts **44** extend between the mating end **30** and the mounting end **32**. In the illustrative embodiment shown, the ground contacts **44** are electrically connected to ground shields or plates **46** that extend between the mating end **30** and the mounting end **32**.

(21) In the illustrative embodiment shown in FIGS. **1** through **5**, the ground shields or plates **46** are positioned or stitched between the columns of contacts **40**. As shown in FIG. **5**, the ground shields **46** have mating ends **48** and mounting ends **50**. The mating ends **50** are positioned proximate the mating end **30** of the connector **12** and the mounting ends **52** are positioned proximate the mounting end **32** of the connector **12**. The ground shields **46** are formed of conductive material, such as, but not limited to, metal or plated plastic.

(22) As shown in FIG. **5**, the ground shields **46** have planar members **52** with surfaces **54**. The surfaces **54** are configured to engage the ground pathways or traces **28** on the backside of the circuit boards **22**. Voids **56** are provided in the planar members **54**. In the illustrative embodiment shown, the voids **56** are raised portions **58** with transition sections **60** and bottom surfaces **62**. The transition sections **60** extend at an angle between the planar surfaces **52** and the bottom surfaces **62**. The bottom surfaces **62** extend in a plane which is removed and parallel to the plane of the planar members **52**. In the embodiment shown, the raised portions **58** are embossed from the planar members **52**. However, other configurations of the raised portions **58** may be used.

(23) Securing openings **64** extend through the transition sections **60** of the raised portions **58**. The securing openings **64** are positioned proximate to the mating ends **48** of the ground shields **46**.

(24) Mounting members **66** extend from the mounting ends **50** of the ground shields **46**. The mounting members **66** have ground contact engagement portions **68** which extend approximately perpendicular to the plane of the planar members **52**. The ground contact engagement portions **68** have openings **70** for receiving the ground contacts **44** therethrough.

(25) In the illustrative embodiment shown in FIGS. **1** through **5**, ground contacts **72** are provided at the ends of the planar members **52**. The ground contacts **72** provide additional ground connections between the ground shields **46** and the backplane (not shown). The ground contacts **72** extend approximately perpendicular to the plane of the planar members **52** and have securing portions **74** which extend from the mounting end **50**. In the embodiment shown, the securing portions **74** are eye of the needle compliant members, but other types of securing portions **74** may be used.

Retention tabs **75** may also be provided on the ground contacts **72**. The retention tabs **75** cooperate with the housing **34** of the backplane connector **12** to secure the ground shields **46** in position.

(26) When the ground shields **46** are properly inserted into ground shield receiving slots **76** (FIG. **3**) of the backplane connector **12**, the raised portions **58** of a respective ground shield **46** is positioned in line with the ground contacts **44** of an adjacent row of contacts **40**. Consequently, the bottom surfaces **62** of the raised portions **58** are spaced from, and do not engage, the signal contacts **42**. The ground contacts **44** and the ground shields **46** peripherally surround pairs of signal contacts **42** to provide electrical shielding for the pairs of signal contacts **42**. Gaps or spaces, which could allow EMI leakage between pairs of signal contacts **42**, are minimized through or between the ground contacts **44** and the ground shields **46**.

(27) When the ground shields **46** are properly inserted into ground shield receiving slots **76** of the backplane connector **12**, the openings **70** of the ground contact engagement portions **68** of the mounting members **66** are positioned about mounting portions **78** of the ground contacts **44**. The ground contacts **44** and the planar members **52** and the raised portions **54** of the ground shields **46** form cavities **80** (FIG. **3**) around pairs of signal contacts **42**. The cavities **80** may have any shape

depending on the shapes of the ground contacts **44** and the planar members **52** and the raised portions **54** of the ground shields **46**.

(28) The ground shields **46** are retained in position in the ground shield receiving slots **76** of the backplane connector **12** by retention projections **81** of the backplane connector **12**. As shown in FIG. **1**, the retention projections **81** extend through the securing openings **64** to properly position and retain the ground shields **46** in the ground shield receiving slots **76**. In addition, the retention tabs **75** cooperate with the housing **34** of the backplane connector **12** to secure the ground shields **46** in position.

(29) Upon positioning the backplane connector **12** on the backplane, mounting portions **82** of the signal contacts **42** and the mounting portions **78** of the ground contacts **44** are positioned in openings of the backplane. As the contact engagement portions **68** of the mounting members **66** are positioned about mounting portions **78** of the ground contacts **44**, the contact engagement portions **68** are electrically coupled to the ground traces of the backplane.

(30) Upon mating of the daughtercard connector **14** to the backplane connector **12**, the circuit boards **22** of the daughter card connector **14** are positioned in circuit board receiving slots **84** of the backplane connector **12**. In this position, the signal contacts **42** physically and electrically engage the signal conductive pads **24** of the signal traces of the circuit boards **22**. The ground contacts **44** are also placed in physical and electrical engagement with the ground conductive pads **26** of the ground traces of the circuit boards **22**. In addition, the surfaces **54** of the ground shields **46** are electrically coupled to the ground traces **28** of the circuit boards **22**.

(31) When fully assembled, the contact engagement portions **68** are electrically coupled to the ground traces of the backplane and the surfaces **54** of the ground shields **46** are electrically coupled to the ground traces **28** of the circuit boards **22**, thereby providing an electrical pathway from the circuit boards **22**, across the ground shields **46** to the backplane. Due to the configuration of the components, the electrical coupling may occur due to physical contact, capacitive coupling or both.

(32) In various embodiments, the contact engagement portions **68** are in physical contact with the ground traces of the backplane, thereby allowing the ground current to flow across the physical contact points. Similarly, the surfaces **54** of the ground shields **46** are in physical contact with the ground traces **28** of the circuit boards **22**, thereby allowing the ground current to flow across the physical contact points.

(33) In other embodiments, the contact engagement portions **68** may be proximate to, but physically separated from the ground traces of the backplane due to tolerances, warpage or other factors. Similarly, the surfaces **54** of the ground shields **46** may be proximate to, but physically separated from the ground traces **28** of the circuit boards **22** due to tolerances, warpage or other factors. In these instances, the surface areas of either of the contact engagement portions **68** and the ground traces of the backplane or the surfaces **54** of the ground shields **46** and the ground traces **28** of the circuit boards **22** are of sufficient size to allow the ground current to flow across by capacitive coupling, with no physical contact. In so doing, the ground shield **46** and the copper in the daughter card or backplane short together to make electrical contact at high frequencies.

(34) If connections from the ground shield **46** to both the daughter card and the backplane are made by capacitive coupling, the ground shield **46** would be considered a floating ground. The capacitive coupling can occur through air, solder mask or through metal to metal connections with little or no normal force.

(35) With the connector **12** properly assembled, the ground contacts **44** and the ground shields **46** extend about the periphery of the pairs of signal contacts **42** and surround the pairs of signal contacts **42** to provide electrical shielding for the pairs of signal contacts **42** and to control cross talk, signal radiation, impedance or other electrical characteristics.

(36) FIGS. **6** through **8** illustrate an alternate grounding shield **146**. The ground shields or plates **146** are positioned or stitched between the columns of contacts **140**. The ground shields **146** are formed of conductive material, such as, but not limited to, metal or plated plastic.

(37) The ground shields **146** have planar members **152** with surfaces **154**. The surfaces **154** are configured to engage the ground pathways or traces **28** on the backside of the circuit boards **22**. Voids **156** are provided in the planar members **154**. In the illustrative embodiment shown, the voids **156** are windows or openings **159**. In the embodiment shown, the openings **159** are stamped from the planar members **152**. However, other configurations of the openings **159** may be used.

(38) Mounting members **166** extend from the mounting ends **150** of the ground shields **146**. The mounting members **166** have ground contact engagement portions **168** which extend approximately perpendicular to the plane of the planar members **152**. The ground contact engagement portions **168** have openings **170** for receiving the ground contacts **144** therethrough.

(39) When the ground shields **146** are properly inserted into ground shield receiving slots **176** of the backplane connector **112**, the openings **159** of a respective ground shield **146** is positioned in line with the ground contacts **144** of an adjacent row of contacts **140**. The ground contacts **144** and the ground shields **146** peripherally surround pairs of signal contacts **142** to provide electrical shielding for the pairs of signal contacts **142**. Gaps or spaces, which could allow EMI leakage between pairs of signal contacts **142**, are minimized through or between the ground contacts **144** and the ground shields **146**.

(40) When the ground shields **146** are properly inserted into ground shield receiving slots **176** of the backplane connector **112**, the openings **170** of the ground contact engagement portions **168** of the mounting members **166** are positioned about mounting portions **178** of the ground contacts **144**. The ground contacts **144** and the planar members **152** and the openings **159** of the ground shields **146** form cavities **180** (FIG. 7) around pairs of signal contacts **142**.

(41) Upon positioning the backplane connector **112** on the backplane, mounting portions **182** of the signal contacts **142** and the mounting portions **178** of the ground contacts **144** are positioned in openings of the backplane. As the contact engagement portions **168** of the mounting members **166** are positioned about mounting portions **178** of the ground contacts **144**, the contact engagement portions **168** are electrically coupled to the ground traces of the backplane.

(42) Upon mating of the daughtercard connector **14** to the backplane connector **112**, the circuit boards **22** of the daughter card connector **14** are positioned in circuit board receiving slots **184** of the backplane connector **112**. In this position, the signal contacts **142** physically and electrically engage the signal conductive pads **24** of the signal traces of the circuit boards **22**. The ground contacts **144** are also placed in physical and electrical engagement with the ground conductive pads **26** of the ground traces of the circuit boards **122**. In addition, the surfaces **154** of the ground shields **146** are electrically coupled to the ground traces **128** of the circuit boards **122**.

(43) When fully assembled, the contact engagement portions **168** are electrically coupled to the ground traces of the backplane and the surfaces **154** of the ground shields **146** are electrically coupled to the ground traces **28** of the circuit boards **122**, thereby providing an electrical pathway from the circuit boards **122**, across the ground shields **146** to the backplane. Due to the configuration of the components, the electrical coupling may occur due to physical contact, capacitive coupling or both.

(44) In various embodiments, the contact engagement portions **168** are in physical contact with the ground traces of the backplane, thereby allowing the ground current to flow across the physical contact points. Similarly, the surfaces **154** of the ground shields **146** are in physical contact with the ground traces **128** of the circuit boards **122**, thereby allowing the ground current to flow across the physical contact points.

(45) In other embodiments, the contact engagement portions **168** may be proximate to, but physically separated from the ground traces of the backplane due to tolerances, warpage or other factors. Similarly, the surfaces **154** of the ground shields **146** may be proximate to, but physically separated from the ground traces **28** of the circuit boards **22** due to tolerances, warpage or other factors. In these instances, the surface areas of either of the contact engagement portions **168** and the ground traces of the backplane or the surfaces **154** of the ground shields **146** and the ground

traces **28** of the circuit boards **22** are of sufficient size to allow the ground current to flow across by capacitive coupling, with no physical contact. In so doing, the ground shield **146** and the copper in the daughter card or backplane short together to make electrical contact at high frequencies.

(46) If connections from the ground shield **146** to both the daughter card and the backplane are made by capacitive coupling, the ground shield **146** would be considered a floating ground. The capacitive coupling can occur through air, solder mask or through metal to metal connections with little or no normal force.

(47) With the connector **112** properly assembled, the ground contacts **144** and the ground shields **146** extend about the periphery of the pairs of signal contacts **142** and surround the pairs of signal contacts **142** to provide electrical shielding for the pairs of signal contacts **142** and to control cross talk, signal radiation, impedance or other electrical characteristics.

(48) FIGS. **9** and **10** illustrate another alternate grounding shield **246**. Ground shields **246** are similar to ground shields **46**, with the exception of the configuration of the voids **256** or raised portions **258**. The ground shields **246** have planar members **252** with surfaces **254**. The voids **256** or raised portions **258** have transition sections **260** and bottom surfaces **262**. The transition sections **260** extend at an angle between the planar surfaces **252** and the bottom surfaces **262**. The transition sections **260** are provided on three sides. Open sides **261** are provided proximate the mating ends **248** of the grounding shields **246**. The bottom surfaces **262** extend in a plane which is removed and parallel to the plane of the planar members **252**. In the embodiment shown, the raised portions **258** are embossed from the planar members **252**. However, other configurations of the raised portions **258** may be used.

(49) The open sides have securing projections **264** which extend from the bottom surfaces **262**. The securing projections **264** engage securing members **235** of the housing **234** of the backplane connector **212** to retain the grounding shield **246** in position.

(50) Mounting members **266** extend from the mounting ends **250** of the ground shields **246**. The mounting members **266** have ground contact engagement portions **268** which extend approximately perpendicular to the plane of the planar members **252**. The ground contact engagement portions **268** have openings **270** for receiving the ground contacts **244** therethrough.

(51) The operation and function of the grounding shield **246** is similar to the described for the ground shield **46**.

(52) While the invention has been described with reference to a preferred embodiment, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the spirit and scope of the invention as defined in the accompanying claims. One skilled in the art will appreciate that the invention may be used with many modifications of structure, arrangement, proportions, sizes, materials and components and otherwise used in the practice of the invention, which are particularly adapted to specific environments and operative requirements without departing from the principles of the present invention. The presently disclosed embodiments are therefore to be considered in all respects as illustrative and not restrictive, the scope of the invention being defined by the appended claims, and not limited to the foregoing description or embodiments.

Claims

1. An electrical connector which controls cross talk and signal radiation, the electrical connector comprising: a housing having a mating end and an oppositely facing mounting end, the mating end extending parallel to the mounting end; columns of contacts having signal contacts and ground contacts extending from proximate the mating end through the oppositely facing mounting end; ground shields positioned between the columns of contacts, the ground shields contacts extending from proximate the mating end through the oppositely facing mounting end, the ground shields comprising: planar members having mating ends provided proximate the mating end of the housing

and mounting ends provided proximate the oppositely facing mounting end of the housing, the planar members having surfaces which are configured to be electrically coupled to ground members on a mating connector; voids provided in the planar members; and mounting members extending from the mounting ends of the planar members, the mounting members having contact engagement portions configured to cooperate with the ground contacts provided in the columns of contacts, the mounting members positioned in line with the voids of the planar members.

2. The electrical connector as recited in claim 1, wherein the voids in the planar members are raised portions with bottom surfaces positioned in a second plane which is removed from and parallel to a first plane of the planar members.
3. The electrical connector as recited in claim 2, wherein a transition sections extend between the planar members and the bottom surfaces of the raised portions.
4. The electrical connector as recited in claim 3, wherein the raised portions are embossed from the planar members.
5. The electrical connector as recited in claim 3, wherein securing openings extend through the raised portions, the securing openings receiving securing projections which extend from the housing of the electrical connector.
6. The electrical connector as recited in claim 1, wherein the voids in the planar members are openings in the planar members.
7. The electrical connector as recited in claim 1, wherein the planar members are electrically coupled to the ground members on the mating connector by direct physical contact.
8. The electrical connector as recited in claim 1, wherein the planar members are electrically coupled to the ground members on the mating connector by capacitive coupling.
9. The electrical connector as recited in claim 1, wherein the mounting members are approximately perpendicular to the planar members.
10. The electrical connector as recited in claim 9, wherein the mounting members have contact receiving openings through which the ground contacts extend.
11. The electrical connector as recited in claim 9, wherein the mounting members are electrically coupled to a mating circuit board by direct physical contact.
12. The electrical connector as recited in claim 9, wherein the mounting members are electrically coupled to a mating circuit board by capacitive coupling.
13. The electrical connector as recited in claim 1, wherein edge ground contacts extend from the planar members.
14. The electrical connector as recited in claim 13, wherein the edge ground contacts are approximately perpendicular to the planar members.
15. The electrical connector as recited in claim 14, wherein the edge ground contacts have deformable mounting portions which are configured to be positioned in through holes of a mating circuit board.
16. The electrical connector as recited in claim 15, wherein the edge ground contacts have retention tabs which extend in an opposite direction as the deformable mounting portions.
17. The electrical connector as recited in claim 1, wherein the ground contacts and the ground shields peripherally surround pairs of the signal contacts to provide electrical shielding for the pairs of the signal contacts.
